Methodological Simulation Design: Stress-Testing GBTM for Recurrent Psychiatric Count Outcomes Under Observation Loss and Administrative Censoring

Wandering Data Ghost #2

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Project Framing

This independent methodological framework was designed to address an analytical bottleneck observed in an ongoing psychiatric epidemiology study of PMD. The practical concern was straightforward: group-based trajectory modeling (GBTM) may identify meaningful trajectory patterns in a recurrent psychiatric count outcome, but its class assignments can be distorted when later outcome events disappear from the observed record or follow-up is truncated by the study end date.

This concern arises from several realistic features of psychiatric follow-up data. Depending on how the outcome is operationalized, later recorded events may disappear from the observed record for substantively different reasons: treatment may succeed after only a small number of recorded events; clinically relevant events may continue but no longer receive formal diagnosis; care may continue through medication use even when no new diagnosis is recorded; or recording may be interrupted by death, severe deterioration, diagnostic migration, or emigration. Separately, some individuals will contribute only short follow-up because they enter the cohort close to the administrative study end. In the observed data, these mechanisms can all flatten or truncate trajectories in ways that are easy to misread as genuinely low burden, remission, or a distinct low-intensity trajectory pattern.

The simulation is therefore intended as a **methodological stress test**, not as proof that a specific class structure must exist in the empirical cohort. Its purpose is to ask a narrower and more answerable question: *how much observation loss and administrative censoring can*

GBTM tolerate before class recovery, trajectory estimation, and classification quality begin to fail?

Although empirically anchored, this design still provides a meaningful quantification of "misclassification" in a fundamentally data-driven method. Here, "misclassification" is used in an epidemiologic and operational sense to describe discordance between recovered classes and the known simulated truth, rather than as an intrinsic error construct of GBTM itself. This is important because GBTM does not recover a directly observed phenotype; it constructs latent trajectory classes from the observed data pattern itself, and is therefore sensitive to non-random data truncation. By quantifying the extent to which observation loss and censoring distort class assignment, the simulation functions as a formal sensitivity stress test and provides an interpretable estimate of the likely magnitude of "misclassification" bias.

Study Aims

Primary aim. To quantify how increasing levels of observation loss and administrative censoring affect class recovery, trajectory estimation, and internal classification quality in GBTM for a single recurrent psychiatric count outcome following PMD.

Secondary aims.

1. To evaluate robustness under alternative operationalizations of the count outcome, such as diagnosed episodes or standardized medication dispensings.
2. To estimate bias in recovered class proportions and mean trajectories relative to the known data-generating truth.
3. To identify practical stress points at which attrition rate, average posterior probability (AvePP), and odds of correct classification (OCC) indicate unacceptable model performance.

1 Methods

1.1 Conceptualization and Motivation

The target setting is a longitudinal PMD cohort with a repeated count outcome for one psychiatric condition of interest. The count may be operationalized, for example, as the number of diagnosed episodes within a follow-up interval or as a standardized count of medication dispensings over that interval. In real-world records, these counts can be degraded in two different ways.

Observation loss is used here as a broad umbrella term for mechanisms that cause later episodes to disappear from the observed trajectory, even though the underlying clinical course or its interpretation may differ across patients. This umbrella may include true treatment success, episodes that continue to occur but are no longer formally diagnosed, and other competing events that interrupt recording, such as death, severe deterioration, diagnostic migration, or emigration. **Administrative censoring** is different: follow-up ends because of study design rather than because the patient's observed trajectory changes for one of these substantive reasons. For example, a patient who enters the cohort shortly before the study end date simply has no observable data beyond that point.

Both mechanisms flatten observed trajectories, but they do so for different reasons. Because the true latent class membership is unknown in empirical data, the resulting "misclassification" bias cannot be measured directly. A Monte Carlo simulation with known ground truth is therefore the appropriate design for evaluating robustness.

1.2 Simulation Architecture

Each simulation replicate will start from a fully observed synthetic cohort and will then impose sequential data degradation before fitting the planned single-outcome GBTM. The baseline scenario is intentionally constrained so that the effect of data corruption can be isolated without creating an infeasibly large scenario space.

- **Sample size:** $N = 50{,}000$.
- **Outcome:** Repeated interval-specific counts for a single psychiatric outcome, defined for example as diagnosed episodes or standardized medication dispensings.
- **Latent classes:** Three classes representing high-, intermediate-, and low-burden recurrent trajectories for that outcome.
- **Primary count model:** Poisson outcomes with quadratic mean trajectories over time.
- **Primary stress parameters:** severity of observation loss and the distribution of administrative censoring times.

Let $i = 1, \ldots, N$ index individuals, $t = 1, \ldots, T$ index follow-up intervals, and $g = 1, \ldots, K$ index latent classes. Each individual is assigned a true class Z_i with class probabilities $\pi_1, \ldots, \pi_K$. Conditional on class membership,

$$Y_{it}^{\text{true}} \mid Z_i = g \sim \text{Poisson}(\lambda_{gt}),$$

with

$$\log(\lambda_{gt}) = \beta_{0g} + \beta_{1g} t + \beta_{2g} t^2.$$

This formulation keeps the primary data-generating mechanism interpretable and close to the planned empirical single-outcome model.

1.3 Empirically Anchored Ground Truth

The simulation is empirically anchored rather than purely hypothetical. The number of latent classes, approximate class proportions, time scale, and plausible trajectory shapes will be informed by the best-fitting single-outcome GBTM applied to the selected psychiatric count outcome in the real PMD cohort.

However, the empirical coefficients will not be copied directly into the simulation truth. Real-world trajectories are already attenuated by observation loss and censoring. The simulation will therefore use empirically informed but deliberately adjusted coefficients to construct a cleaner baseline trajectory structure. This adjusted structure serves as the ground truth against which model robustness can be evaluated.

1.4 Observation Loss and Administrative Censoring

Observation loss will be modeled as a simplified reduction of several substantively different mechanisms into one operational rule. In the illustrative primary scenario, once a trajectory has accumulated three observed events for the selected outcome, all later observed values will be set to zero. This can be understood as a coarse approximation to settings in which patients experience no more than three recorded events before later information disappears from the observed series, whether because treatment succeeds, subsequent events are not formally diagnosed, or another competing event prevents further recording. Sensitivity analyses can later vary the trigger threshold or onset time, and the same logic can be applied whether the observed count is based on diagnosed episodes or standardized medication dispensings.

Administrative censoring will be modeled separately. Each individual will receive a censoring time C_i drawn from a right-skewed distribution calibrated to the empirical enrollment pattern and fixed study end date. All outcomes after C_i will be treated as unobserved, not as zeros.

The order of corruption is important. Observation loss will be imposed first, followed by administrative censoring. This sequence mirrors the substantive logic of the data: observation loss can arise within nominal follow-up, whereas administrative censoring truncates the remaining observation window.

1.5 Model Fitting and Evaluation Metrics

For each corrupted dataset, the planned GBTM will be fitted using the same single-outcome specification and time functional form as in the empirical analysis. In the primary analysis, the number of fitted classes will be fixed to the true value so that the effect of data corruption can be examined separately from the problem of class enumeration. A sensitivity analysis may allow the number of classes to be reselected by a standard criterion such as the Bayesian information criterion.

Because latent class labels are exchangeable, recovered classes will be relabeled to the true classes using a one-to-one matching rule based on maximum overlap. For example, if the recovered class with the largest share of true high-burden individuals is class 2, that class will be relabeled as the recovered high-burden class.

The primary performance measure will be the **attrition rate**, defined here as class-specific "misclassification" in the epidemiologic sense:

$$\text{Attrition Rate}(g) = 1 - \frac{n(\hat{Z}_i = g \mid Z_i = g)}{n(Z_i = g)}.$$

Secondary performance measures will include:

- bias in recovered class proportions;
- bias in predicted mean trajectories or trajectory parameters;
- average posterior probability (AvePP), which reflects how confidently individuals are assigned to a recovered class;
- odds of correct classification (OCC), which reflects how strongly the model favors correct over incorrect assignment.

AvePP and OCC will be interpreted as supporting indicators of internal adequacy, whereas attrition rate provides the direct comparison with known truth.

This distinction has practical importance beyond model diagnostics. In psychiatric epidemiology, recovered trajectory classes are often used in downstream analyses as exposures, outcomes, or stratifying variables. Quantifying the extent of class "misclassification" therefore helps characterize the likely direction and magnitude of downstream bias, which in turn strengthens the robustness and interpretability of subsequent association or causal analyses.

1.6 Monte Carlo Execution and Scenario Grid

For each scenario, the full generate-corrupt-fit-evaluate procedure will be repeated for 1,000 simulation replicates with fixed random seeds. The primary scenario grid will vary only the key stress parameters:

- the severity of observation loss, defined by the trigger threshold or onset time;

- the distribution of administrative censoring times.

Sensitivity analyses may additionally vary sample size, class proportions, the operationalization of the count outcome, and mild forms of model misspecification such as stronger overlap between classes. Across scenarios, the simulation will summarize the mean, empirical standard deviation, and distribution of attrition rate, AvePP, and OCC in order to identify the point at which GBTM for a single recurrent count outcome ceases to perform adequately.

1.7 Interpretive Scope

In the primary analysis, observation loss is intentionally defined broadly and operationally rather than etiologically. The simplified cutoff rule is not intended to distinguish whether later events disappear because of treatment success, lack of formal diagnosis, or another competing event; it is intended only to stress-test how GBTM behaves when the observed trajectory is truncated within nominal follow-up. If needed, later sensitivity analyses can separate these mechanisms and encode them differently in the latent event process.

Illustrative Pilot Result

The current pilot code produced one illustrative realization under the simplified degradation rule described above, namely that observed values were set to zero after three recorded events and then subject to administrative censoring. This result is included only to demonstrate the type of distortion that the planned simulation is intended to quantify; it should not be interpreted as a stable estimate, because it depends strongly on a single stochastic realization and on the present stylized degradation assumptions.

In this pilot realization, class-specific attrition rates were low for the low-burden class (approximately 0.09) but substantially higher for the intermediate- and high-burden classes (approximately 0.54 and 0.72, respectively). Among individuals in the true intermediate-burden class, about 50.5% were reassigned to the recovered low-burden class, 45.7% remained in the recovered intermediate-burden class, and 3.8% were reassigned to the recovered high-burden class. Under the current pilot setting, the dominant error pattern therefore appears to be downward reassignment from more active trajectories into lower-burden classes rather than symmetric misclassification across all classes.

Figure 1 shows the corresponding trajectory recovery pattern. When the same model was fitted to complete data, the recovered trajectories closely tracked the theoretical class-specific means. After observation loss and administrative censoring were imposed, the recovered low-burden trajectory remained broadly similar to the truth, whereas the intermediate- and high-burden trajectories were pulled sharply downward after early follow-up and approached zero over much of the remaining observation window. Under the current cutoff-based degradation rule, this pattern is consistent with the possibility that higher-burden subjects reach the observation-loss threshold earlier and therefore lose more of their later recorded events. This interpretation should remain provisional and specific to the present pilot code until it is evaluated across repeated Monte Carlo runs and alternative degradation mechanisms.

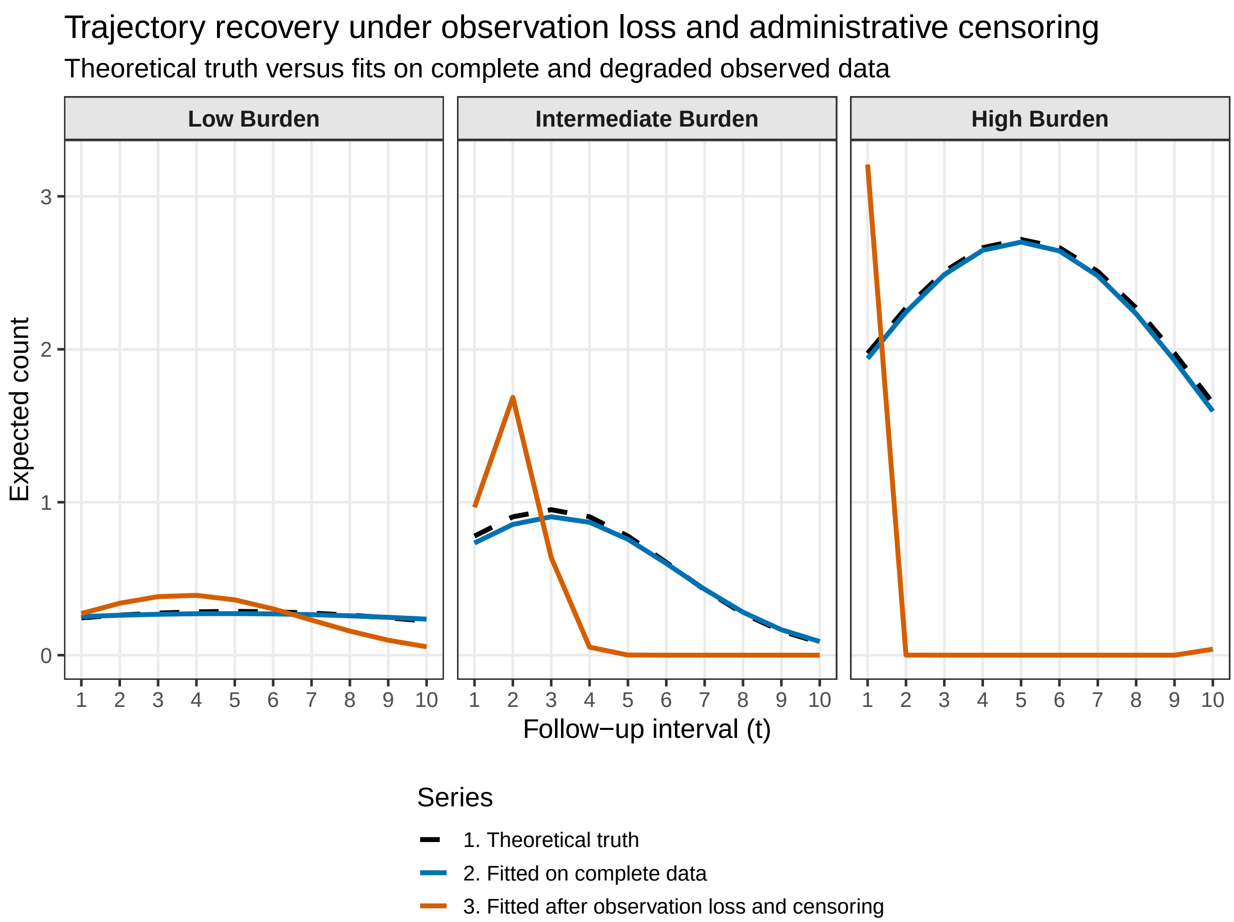

Figure 1: Illustrative trajectory recovery from the current pilot code under the simplified observation-loss rule and administrative censoring. The dashed black line shows the theoretical class-specific mean, the blue line shows the fit on complete data, and the orange line shows the fit after data degradation.